

Release History

Notable changes between versions

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ABSTRACT

This document records the user-visible changes to the project. The format is loosely based on the Keep a Changelog convention; versioning follows Semantic Versioning.

1. Version 0.3.0 — May 2026

Added

- Semantic search using TF-IDF vector embeddings.
- Search analytics tracking with query history and Prometheus metrics.
- Query autocomplete API with real-time suggestions.
- Related queries feature to discover similar search terms.
- Faceted search with filters (domain, source, score) and sorting options.
- Search history & bookmarks with LocalStorage and theme toggle.
- Document management API (add/remove documents).
- Batch search API (/api/batch-search) for multiple queries.
- Export results as JSON or CSV (/api/export).
- Enhanced health endpoints (/healthz, /readyz, /metrics).
- Incremental indexing with SQLite persistence backend.
- Updated web UI with autocomplete, bookmarks, and dark/light theme.
- New API endpoints: /api/semantic-search, /api/autocomplete, /api/analytics, /api/top-queries, /api/related-queries.
- Updated version to 0.3.0 with all new features.

Changed

- Repository URL updated to <https://github.com/Hamza35779/NLP-AI-Search-Engine>.
- All author references updated to Hamza Abdul Karim · F23607046.
- Enhanced README with comprehensive feature list and new sections.
- Rebuilt all PDF documentation with updated content.

2. Version 0.2.0 — May 2026

Added

- Brand identity: SVG logo and favicon.
- Modular CSS bundle (base, components, home, results).
- /about page describing the project and API.
- /healthz liveness endpoint.
- Five black-and-white SVG architecture diagrams.
- Five reference PDFs replacing the previous markdown docs.

Changed

- Polished search form, results page, and result cards.

- Sidebar on the results page now shows the parsed query, index stats, and operator help.
- Redesigned hero with a feature grid and a how-it-works strip.

3. Version 0.1.0 — May 2026

Added

- Core NLP preprocessing.
- Inverted index with positions and per-document statistics.
- TF-IDF (cosine) and Okapi BM25 ranking models.
- Query parser supporting +must, -must-not, "phrase", and boolean operators.
- Snippet generator with highlighting.
- Damerau-Levenshtein spell suggestions.
- IR metric helpers: precision/recall@k, MAP, MRR, nDCG@k.
- CLI with one-shot and REPL modes.
- Flask web UI with HTML pages and a JSON API.
- Pytest suite with 25 tests.
- Sample corpus of seven NLP-themed documents.